

JEE MAIN REVISION MASTER: HEAT AND THERMODYNAMICS 2026 3 QUESTIONS 5% WEIGHTAGE

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 28 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Kcet Heat And Thermodynamics - Core Application	28 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Heat And Thermodynamics - Core Application (Very High Yield — 28 PYQs)

Core Principle: Key problem pattern in Kcet Heat And Thermodynamics.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Following are statements of a few processes taking place in nature. I. Free expansion of a gas II. The combustion of a mixture of petrol and..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "In thermodynamic processes, which of the following statements is not true?..." [Ans: N/A]
- [2026 Shift] Q#3 (MCQ): "The graph of pressure P and volume V of 1 mole of an ideal gas at constant temperature is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Following are statements of a few processes taking place in nature. I. Free expansion of a gas II. The combustion of a mixture of petrol and air ignited by a spark III. The leaking of gas from the kitchen cylinder. IV. The transfer of heat from one heated part of a liquid to the other colder part. Which amongst these processes are irreversible in nature?

- (A) I and II
- (B) I and III
- (C) II, III and IV
- (D) I, II, III and IV

Correct Answer: None

Solution Analysis: All the given processes are irreversible in nature. Free expansion of a gas is irreversible because the gas spreads into the available space on its own. It cannot return to its original volume without some external work being done. Combustion of petrol and air is also irreversible. Once petrol burns, it forms new substances like carbon dioxide, water vapour, etc. These products cannot change back into petrol and air by simply reversing the process. Leaking of gas from a kitchen cylinder is irreversible because the gas moves from high pressure inside the cylinder to lower pressure outside. It w